
Exploring the Use of State-of-the-Art Image Captioning for Predicting Text Prompts in Stable Diffusion

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Abstract

This paper proposes to explore the use of state-of-the-art image captioning methods for the task of predicting prompts from Stable Diffusion, which is a popular text-to-image model. The authors aim to demonstrate the effectiveness of using Stable Diffusion 2.0 to generate image and text data for fine-tuning. The proposed methods use prompt-image pairs to fine-tune different state-of-the-art image captioning models, including Contrastive Language-Image Pre-Training (CLIP) model, and Bootstrapping Language-Image Pre-training (BLIP) model, Generative Region-to-text Transformer for Object Understanding (GRiT) for dense image captioning. These models will invert the diffusion process and obtain prompts used to create images. Further evaluation will be done for traditional image captioning benchmarks.

1 Introduction

Text-to-image models based on diffusion, such as Stable Diffusion [21] and DALL-E-2 [19], have had a recent surge in popularity by enabling high-quality image generation from natural language text prompts. Originating from text-to-text generation, The idea of text prompting, using natural language to guide language-based models to generate desired outputs, has spurred an entirely new field of "prompt engineering" [20].

To help understand how prompts and images are related, Google, LLC has recently created a Kaggle Competition to "Deduce the prompts that generated our "highly detailed, sharp focus, illustration, 3d renders of majestic, epic" images" [4] from Stable Diffusion 2.0. The task is to use images generated by Stable Diffusion 2.0 from a variety of non-disclosed text prompts and methods and predict the text prompts used to generate the images.

Image-to-text, more commonly termed image captioning, has been an active source of research. Much recent work for image captioning involves first using a pre-trained embedding model to encode replaced text and image into the same embedding space [18, 13] and using it for the downstream task of image captioning.

We propose to explore the use of these state-of-the-art image captioning methods for the task of predicting the prompts from Stable Diffusion. At the same time, we will explore how well training the image captioning model (decoding text-image embeddings from images into text) perform for normal image captioning task, demonstrating the effectiveness of using Stable Diffusion to generate image and text data more fine-tuning.

2 Related Work

The process of generating text to describe an image is a task that combines NLP and computer vision techniques. The previous approach to achieve this task is combining the CNN model and RNN model. [7] Thanks to the large-scale pre-trained foundation models, the new state-of-art models are able to be more efficient and effective.

2.1 mPLUG

The mPLUG model, introduced in May 2022, provides a special approach to tackle the problem of low computational efficiency and information asymmetry.[12] The conventional cross-models use direct alignment image and text representation and implement full self-attention or cross-attention on the concatenation of the visual sequences and text sequences. Different from the previous method, mPLUG introduces novel cross-modal skip-connections, by creating inter-layer shortcuts that skip a certain number of full self-attention layers on the vision model to produce better and faster results. The model achieves 4 times speeding up compare to other cross-modal fusion networks. Show in figure 1



Figure 1: mPLUG Run time

The mPLUG model is pre-trained on multiple datasets, including COCO[14], Visual Genome[11], SBU Caption Dataset[16], CC3M[5] and CC12M[5]. The model result is evaluated on COCO[14] and Visual Question Answering test set[3] on cross-entropy loss. The result shows that in the Visual Question Answering test set mPLUG performs better than most of the image captioning models by at least 1 percent. mPLUG also shows better performance in the cross-entropy loss of the COCO dataset and the NoCaps dataset.

2.2 BLIP

Another state-of-art model in Vision-Text generation is Bootstrapping Language-Image Pre-training(BLIP) [13], which was introduced in Feb 2022. Lots of Vision-Language Pretraining models are trained on noisy image-text data collected from the web, which causes low performance. To prevent that, BLIP bootstraps the noisy caption data by filtering noisy data using synthetic captions generated by a captioner. To perform different tasks on image text, BLIP uses a multimodal encoder-decoder. BLIP uses ViT(visual transformer) as its image encoder to receive input of images, and implements a Unimodal encoder, an Image-grounded text encoder, and an Image-grounded text decoder to perform various tasks. The Architecture of BLIP is shown in figure 2.

The BLIP model is pre-trained on multiple datasets, including COCO[14], Visual Genome[11], SBU Caption Dataset[16], CC3M[5], CC12M[5], and LAION[22]. The model result is evaluated on the COCO Caption test dataset[14] and NoCaps validation set[2] using Language Modeling loss. The result shows that the BLIP model performs better than other Vision-Language models in the NoCaps validation data set and COCO dataset. (The mPLUG is not in the comparing group)

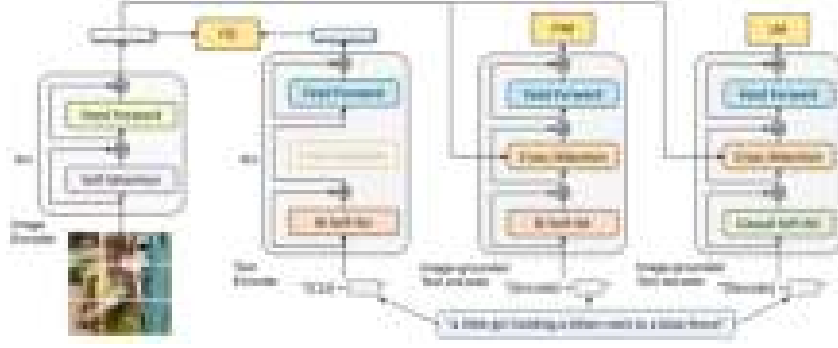


Figure 2: BLIP architecture

3 Proposed Methods

3.1 Dataset

The Kaggle Competition does not provide a dataset, but there are many freely available sources of prompts and images. DiffusionDB is one such dataset that collected many images created using diverse prompts and hyperparameters specified by actual users, resulting in a wide variety of visual styles and content. The dataset contains 14 million images, and a smaller, higher-quality 2 million image subset [24]. However, due to its large size, the subset requires approximately 1.6 TB of storage space. By carefully selecting smaller subsets of the dataset, we can tailor our model training to specific objectives and optimize performance. In addition, we can also generate a dataset of our own using Stable Diffusion 2.0 so that we can future align our goal with Stable Diffusion 2.0 inversion.

3.2 CLIP Image Captioning

We will utilize the Contrastive Language-Image Pre-Training (CLIP) [18] model to fine-tune a model to decode CLIP embeddings of images into text. We will train or fine-tune a model to map CLIP embeddings to become prefixes for a language model, GPT2, enabling CLIP Prefix for Image Captioning as described by Mokady et al. [15]. Using a supervised learning approach, we fine-tune the mapping model and language model on the image-prompt dataset and evaluate its performance using standard metrics. Our iterative process involved adjusting model architecture, hyperparameters, and training data in improving performance. Because Stable Diffusion 2.0 is trained using OpenCLIP [23] as a text decoder, using CLIP to decode images to text should result in accurate prompt descriptions.

3.3 BLIP Image Captioning

We will further fine-tune the Bootstrapping Language-Image Pre-training (BLIP) [13] model to evaluate its performance against other image captioning models. For this purpose, we will omit the Captioning and Filtering (CapFilt) component of BLIP as our dataset comprises labeled images that were generated based on specified prompts. Therefore, there is minimal noise or irrelevant content that requires filtering using CapFilt. We will experiment with various hyperparameters, regularization techniques, and fine-tuning methods to improve the model’s accuracy and generate diverse captions for new images.

3.4 Dense Image Captioning

Most captioning systems are trained to generate a single description of an image. However, prompts, particularly good prompts, are usually highly detailed and long, containing many different descriptions to help guide [1]. The task of dense image captioning attempts to solve this by providing as much detail in a caption as possible.

To do this, we will use previous work that proposes to caption based on different regions of the image and fine-tune the network to generate prompts. This method first performs object detection to

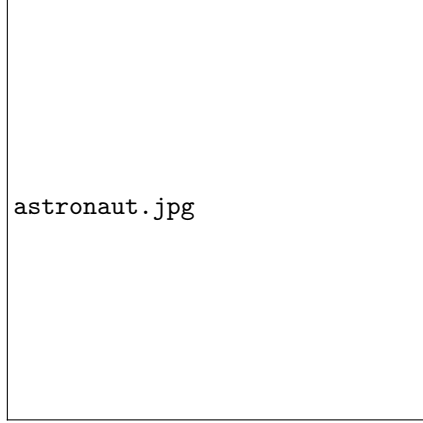


Figure 3: Image found on Lexica, a massive database of stable diffusion images and prompts, and passed into GRiT for dense image captioning. The prompt was: "an astronaut resting on mars in a beach chair, vibrant lighting, elegant, highly detailed, smooth, sharp focus, illustration, beautiful, geometric, trending on artstation, full body, cinematic, artwork by borovikovsky, 8k" [6]

find bounding boxes for objects in the foreground and background. Then a text decoder is trained to generate captions from the images, thus providing descriptions of the entire scene [25]. The hypothesis is that having multiple perspectives on the image help to provide good prompts for stable diffusion.

For this task, however, the base object detection may be too powerful - detecting too many objects in the scene, as seen by Figure 3 where GRiT describes 10 objects (also not that accurately). A foreground/background separator may be all we need to describe the image prompt. We will also fine-tune the text decoder for our specific image-to-prompt task.

3.5 CLIP Interrogator Baseline

CLIP Interrogator [17] is a tool used by many people for the task of predicting prompts from images and a good baseline for this task. It combines CLIP and BLIP to "optimize text prompts to match a given image" [17] by first creating a caption using BLIP, then, from a pre-collected keyword bank, the top-k keywords are appended to the generated prompt based on the similarity score between CLIP embeddings of the keywords and the image.

3.6 Compute

For most of our computing needs, since we are only fine-tuning models and/or composing different models, free online notebooks should be enough. Google Colab hosts free GPUs with 12-16GB vram, but the limit for these GPUs is variable but in general, they can run for at most 12 hours. Kaggle also provides free P100 GPUs and TPUs with 12-hour execution time for CPU notebook sessions and 9 hours for TPU notebook sessions and a weekly 30 hour quota for GPUs [9]. If the 12-hour execution is not enough, we have access to an NVidia 3080 GPU to train for longer.

If, for some reason, this compute is not enough, since we are fine-tuning models, we can also use the Low-Rank Adaptation of Large Language Models (LoRA) technique that injects "trainable rank decomposition matrices into each layer of the Transformer architecture" to "greatly reduce the number of trainable parameters for downstream" [8].

4 Evaluation

As part of the Kaggle competition, we aim to build a model that can predict the text prompt given a generated image, inverting the typical direction of a generative text-to-image model. To evaluate the prompt similarity, we will submit embeddings of our predicted prompts to the Kaggle competition, which will be scored in a robust manner. This means that even if there are character-level differences

between prompts, they will be considered similar. This approach will assess the effectiveness of our model in understanding the underlying relationships between prompts and the images they generate. [10]

Additionally, We plan to use the models we generated to test them on image captioning tasks using datasets such as COCO captions, COCO, or nocaps-nal-in-domin. We will compare the BLUE indexes to evaluate the effectiveness of our models in generating captions.

5 Timeline and Tasks

We are a team of three people, and we have divided our responsibilities as follows: Jason will focus on building the CLIP image-to-prompt model, Woody will focus on building the BLIP image-to-prompt model, and Jimmy will focus on building the Dense Image image-to-prompt model.

March 11: Data collection and pre-processing.

March 25: AI model selection and design.

April 1: AI model training on a subset of the data.

April 15: Model refinement and hyperparameter tuning.

April 29: Model testing and evaluation. Write-up of results and discussion

May 4: Final editing and review, submission.

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